INNATE AI LTD · CONTRACT DEVELOPER HIRING

Test brief: storefront capture & visualisation

Stage 1 of 3

THE CLIENT AND THE PRODUCT

Our client supplies design-led outdoor planters to restaurants, salons and offices in London, on a purchase or monthly rental basis with seasonal replanting. We are building them a prospecting engine. Before any outreach happens, the client wants to identify independent venues around London — cafes, restaurants, salons, and similar street-facing businesses — with bare or under-dressed frontages, where the client's planters would visibly enhance the entrance. For each one, the system needs to produce a realistic visual of that venue's own doorway, dressed with the client's real products, convincing enough to send to the owner.

You are being tested on the two capabilities at the technical heart of that: getting a usable, well-framed photo of a real venue's frontage (real street view), and compositing the client's actual planters onto it believably and at correct scale. You're building the part that turns a venue into a convincing "what it could look like."

For a sense of the ambition here (not the exact build): https://x.com/everestchris6/status/2043061132911677924 — a similar concept applied to backyard pools, sourced and rendered fully automatically. We're not replicating that build, but it's a useful reference point for the level of polish and realism we want the output to have.

You are expected to use AI coding tools throughout. We are assessing your decisions, not your typing, and every decision in your submission must be one you can defend on a call without notes.

WHAT YOU WILL BUILD

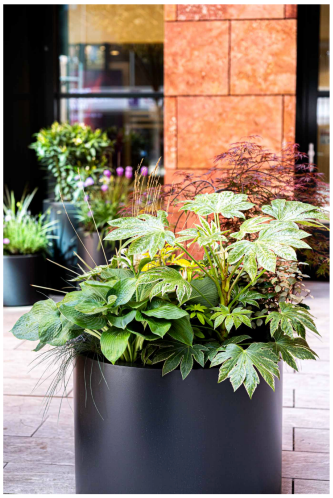
Your submission has three parts.

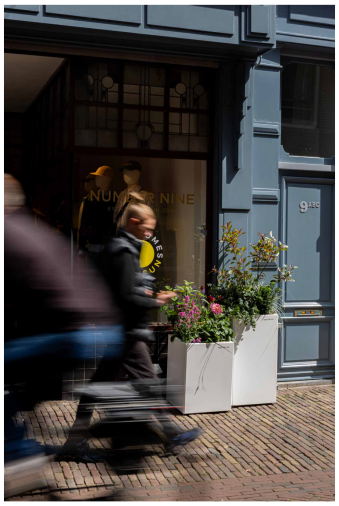
1. A short design note, `design.md`, covering both capabilities below at a depth a second engineer could build from.
2. A working prototype, built end to end on the three venues you choose, that takes a venue and outputs a composited "planters installed" visual.

WHAT YOUR PROTOTYPE NEEDS TO DO

Getting the frontage image. From a venue's location, produce a well-framed image of its actual entrance — not a generic stock photo, the real doorway. Your design note should cover: how you'd derive a usable framing from a location and, where relevant, panorama data (camera heading, field of view); what you do when the nearest available imagery faces the wrong way or doesn't clearly show the entrance; and your written position on the imagery-rights question this raises — you're capturing and reusing a photo of someone's real property at commercial scale, without their involvement at this stage.

Compositing the planters. Take the client's real product photography and place it into the frontage image so it looks like it's actually there — correct scale, correct perspective, believable shadow and grounding, the building and everything else in the shot left unaltered. This must use the client's actual products — three reference photos are provided below — not a generic AI approximation of "some planters."





Your design note should cover: how you'd estimate real-world scale from a reference object in the photo; how you'd keep the products visually faithful to the reference photos rather than letting the model reinterpret them; and your rejection criteria — what makes a generation bad enough that it should never reach a venue owner.

CHOOSING YOUR VENUES

Rather than working from a fixed list, your prototype needs to find roughly three or more candidate venues in London itself — independent cafes, restaurants, salons, or similar street-facing businesses with a bare or under-dressed frontage where the client's planters would visibly improve the entrance. This selection step should be automated, not something you curate by hand: pull a candidate list from a real source (the Google Places API, OpenStreetMap, or another source of your choice), then have your own code — AI-assisted or not — decide which ones actually look like good fits, using whatever signals you can reasonably extract (business type, or a vision-model judgement on the frontage itself, for instance).

In `design.md`, list the venues your prototype selected — name, address, postcode — and explain the logic (or the model prompt/criteria) that picked them, along with anything it rejected and why. At 5,000+ venues a week, nothing in this pipeline can rely on someone manually eyeballing a good candidate or the best framing of its doorway — the same applies to the frontage-image capture that follows. In practice that means your own code needs to decide, unaided, whether a captured image is actually usable, and if Street View coverage is poor or faces the wrong way, fall back to another source — the venue's own website or Google Business photos are a reasonable choice. State your own accept/reject bar for what makes a candidate venue or a framing usable, and don't over-invest in polish here: one or two rejected attempts before falling back is a perfectly good outcome — we're grading that the decision-making is automated, not that it's flawless.

DELIVERABLES

One GitHub repo containing `design.md`, a working link of your prototype (deployed on Vercel, html, etc.) and `README.md`. Attach the repo link.
